

Prince Sharma

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AI/ML engineer, delivering production LLM and RAG pipelines. Built sub-second semantic search, compliance grade claim-audit extraction, and multi-agent summarization services. Fuse deep mathematics with end-to-end MLOps and cross-cloud engineering to convert research breakthroughs into scalable, globally impactful products.

Education

Indian Institute of Technology Madras

Masters of Technology in Industrial Mathematics and Scientific Computing (GPA: 8.27 / 10)

Jul 2024

Chennai, TN

Delhi Technological University

Master of Science in Mathematics (GPA: 8.47 / 10)

Jul 2022

New Delhi, DL

Technical Skills

Programming & MLOps: Python, SQL, Git, FastAPI, Docker, Kubernetes, Jenkins, Github Actions, OpenSearch, AWS.

Tools & Frameworks: LangChain, LangGraph, Google ADK, CrewAI, ElasticSearch, Milvus, Redis, PostgreSQL, MCP.

Certifications: AWS Cloud Practitioner, AWS AI Practitioner, Microsoft Azure AI Fundamentals.

Experience

Software Engineer | Virtusa | Chennai, TN

Aug 2024 - Jul 2026

- Enhanced search suggestion relevance by building a customized Elasticsearch autocomplete pipeline with n-gram indexing and query-time boosting, reducing irrelevant results and improving user query experience.
- Built an agentic CI-ready chatbot testing framework using Playwright, evaluating goal-based conversational scenarios via page trails and chat history to detect regressions and guide prompt, RAG, and system improvements.
- Developed a Slack assistant bot that ingests channel history, stores QA pairs in Elasticsearch, and uses a hybrid search approach to retrieve the most relevant previously answered queries, improving support accuracy.
- Designed an LDA-based topic discovery pipeline over production query logs and exposed topic APIs to the OneFind KMS admin console to drive targeted knowledge-article improvements and synonym updates.
- Built the analytics API backing OneFind's Knowledge Management Service admin console—exposing search/NLP metrics for insight retrieval and experiment analysis within OneFind Search-as-a-Service.
- Designed Search-as-a-Service engine for a chatbot that turns queries into embeddings, serves hot paths in under-10 ms via Redis, persists long-tail knowledge in Elasticsearch, and delivers precise sub-second answers.
- Built and deployed a Learn-to-Rank (LTR) pipeline for the RAG chatbot's retrieval layer, achieving nDCG@10 = 0.92 and enhancing answer relevance.

Data Science Project Intern | PDSVISION | Chennai, TN

Aug 2023 - Apr 2024

- Spearheaded automotive-parts dataset creation with advanced collection, labeling, and augmentation, then trained a hyper-optimized YOLOv8 pipeline achieving 0.991 mAP to enable real-time edge-device part verification.

Data Science Intern | Tata Communications | Chennai, TN

May 2023 - Jul 2023

- Curated and annotated 500+ soccer-match videos, labeling goals, fouls, and corners, hyperparameter-tuned EfficientNet-B0 to boost multi-class accuracy by 15 % and enable near-real-time analytics workflows.

Projects

E-Commerce Product Recommendation System | Python, LangGraph

- Built a LangGraph-powered LLM agent that retrieves Platzi Store data, maintains conversational state, and runs multi-step tasks—cart updates, invoicing—via memory-aware tool chains for personalised product recommendations.

Multi-Agentic Research Agent | Python, LangGraph

- Built a research agent using LangGraph to automate multi-step information retrieval and reasoning across sources, integrating LLMs with tool usage to generate structured, context-aware responses for complex queries.

Awards, Achievements & Leadership

Awards & Recognition: Recognized as Star Performer (H1 FY26) at Virtusa for high-impact GenAI/ML contributions.

Achievements: GATE MA AIR 281, NCC 'C' Certificate, Scouting State Award, 2-time U-19 Hockey State Level champion.

Position of Responsibility: Class Representative of MTech IMSC'24, Battery Commander in EBSB Camp.